



How Advanced Sensors And Actuators Are Enabling High-Tech Smarter Aircraft

In its relatively short existence, the aerospace industry has seen more changes than any other major industry. Some of the recent developments of wireless sensors on aircraft structures and health monitoring are highlighted here

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An initiative is underway to develop and deploy wireless data management systems for aircraft. As part of the WiTNESS programme, the technology will be used for preventive maintenance of fixed-wing aircraft and helicopters. Taking queue from other harsh environment installations, the data controller will optimise diagnostics.

Simulation techniques and lightweight construction are enabling BLDC electric drive systems to achieve a power to weight performance of 5kW/kg compared to the industrial electric motors of 1kW/kg. This innovation is being used for propulsion of E-aircraft systems as well as flight control actuators and autopilots.

Wireless smart sensors

Improved control of aircraft, lower maintenance costs, reduced fuel consumption and increased passenger comfort are major benefits of an aircraft manufacturer's initiative to bring wireless sensor networks to commercial and military aviation. Wireless sensors are expected to reduce the maintenance costs of the aircraft, which account for more than 20% of the operating costs of an airliner.

The concept adds an assortment of sensors, which provide additional data or more precise data on parameters already being monitored. Thermoelectric generators power the wireless sensor nodes, eliminating the need to periodically charge batteries or replace them.

Smart skin technology

Aircrafts will have human-like skin that would enable them to detect injury or damage, as well as the ability to feel the world. The skin would consist of tens of thousands of micro sensors applied to the actual aircraft that will enable it to sense wind speed, temperature, and physical strain movement with more accuracy than current sensor technology.

The smart skin will allow an airplane to monitor its overall health and then report its status before any issues become truly problematic. The tiny sensors that would be used in the technology could be as small as grains of rice or even as small as dust particles at less than 1mm squared.

The proposed solution allows the instal-



Airbus A380 (Courtesy: Airbus Industries)